



EE 515: Nonlinear Stability and Control

Course Code: EE 515

Course Name: Nonlinear Stability and Control

L-T-P-C: 4-0-0-4

Prerequisites: EE 301 Control Systems, EE 509 Linear Dynamical Systems or teachers consent

Intended for: U.G./P.G.

Distribution: Elective

Approval: 9th Senate

Course Contents

- **Linear and nonlinear system behaviours:** Quick recapitulation of linear differential equations, and their solutions. Qualitative properties of nonlinear systems. Existence and uniqueness of solutions to Ordinary differential equations. Linearizations. Hartman-Grobman theorem. Volterra's functional expansions. [10 Lectures]
- **Lyapunov's stability theory:** Notions of stability. Lyapunov's stability theorem. LaSalle's invariance principle. Circle criterion, Popov criterion. Lyapunov-Krasovskii functionals. [16 Lectures]
- **Limit cycles, Bifurcations, Chaos:** Poincaré-Bendixson theorem. Center manifold theory. Simple examples of Bifurcations, and, Chaos. [7 Lectures]
- **Describing functions:** The describing function method, Computing amplitude and frequency of oscillations. [6 Lectures]
- **Nonlinear geometric control:** Lie brackets and controllability. Examples. [3 Lectures]
- **Passivity and Backstepping control:** From Absolute stability to Passivity. Passivity based design. Zero dynamics. $L_g V$ control. Control Lyapunov functions. Recursive designs. Examples. [8 Lectures]
- **Lab exercises:** Experiments on Describing functions, Backstepping controllers etc. for Pendulum and cart system, DC to DC converter, Four tank water level control system etc. [10 Lectures]

Note: This is mainly a theoretical course. There is a small experimental component needing about 10 hours of lab work. This is included to give some practical demonstration of techniques taught, and also because some of our students (especially future M.Tech. students) would be heading to industry after their studies.

Textbooks:

1. H. Khalil, **Nonlinear Systems**, 3rd edition, Pearson, 2014
2. W. J. Terrell, **Stability and Stabilization**, Princeton University press, 2009
3. R. Sepulchre et al., **Constructive Nonlinear Control**, Springer, 1996
4. M. Krstic et al., **Nonlinear and Adaptive Control Design**, Wiley-Blackwell, 1995

Additional References:

1. S. Sastry, **Nonlinear Systems: Analysis, Stability, and Control**, Springer, 1999
2. W. Hahn, **Stability of Motion**, Springer, 1968
3. A. Mees, **Dynamics of feedback systems**, Wiley-Blackwell, 1981
4. A. Isidori, **Nonlinear Control Systems**, volumes 1, 2, Springer, 1995 and 1999
5. A. Blaquiere, **Nonlinear System Analysis**, Academic press, 1966
6. W. Haddad, and V. Chellaboina, **Nonlinear Dynamical Systems and Control**, Princeton University press, 2008
7. V. A. Yakubovich, G. A. Leonov, and, A. Kh. Gel'fand, **Stability Of Stationary Sets In Control Systems With Discontinuous Nonlinearities**, World-scientific, 2004